

"PAPER_{0:D3}" -f [int]# PAPER #83 ■ Primordial Black Hole Mass Distribution via UQFF

Author: Daniel T. Murphy

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Abstract

Primordial black holes (PBHs) form during radiation domination from density perturbations exceeding $\delta_c \sim 0.45$. The UQFF modifies the critical density threshold $\delta_{c\text{UQFF}}$ through the Buoyant vacuum pressure correction and extends PBH survival above the GR threshold mass $M_{\text{threshold}}$ by 3.5%. The evaporation temperature of surviving PBHs is $T_{\text{UQFF}} = 0.99 \cdot T_H$, slightly reducing their current gamma-ray flux. PBHs with $M \sim 10^{15} - 10^{17}$ g may constitute part of dark matter ■ the UQFF predicts the dark matter fraction f_{PBH} is unaffected (the 3.5% mass threshold shift does not move PBHs into or out of the observationally allowed window).

UQFF Discovery: Novel application of UQFF calibration constants ($\gamma = 5.0 \cdot 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. PBH Formation Threshold

Standard GR δ_c

$\delta_c = 0.45$; (\text{Harrison-Zel'dovich radiation era})

UQFF Correction

The UQFF Buoyant mode adds vacuum pressure during the radiation era:

$$P_{\rm UQFF} = P_{\rm radiation} + P_{\rm vacuum} = \frac{\rho_{\rm rad} c^2}{3} + \rho_{\rm vac} c^2 \times [UA]$$

The fractional correction:

$$\frac{\Delta P_{\rm UQFF}}{P_{\rm UQFF}} = \frac{\Delta P}{P} \times \left(1 - \frac{P_{\rm vacuum}}{P_{\rm rad}}\right) = 0.45 \times (1 - [UA] \times z_{\rm formation}^{-4})$$

At $z_{\rm formation} = 106$ (typical PBH epoch): $P_{\rm vacuum}/P_{\rm rad} = 0.0001 \ll 10^{-4}$? negligible.

UQFF dc = 0.45 (unchanged from GR) ■ the formation threshold is not perturbed.

2. Survival Mass Threshold

Standard GR

$$M_{\rm threshold}^{\rm GR} = 5.70 \times 10^{11} \text{ kg} \quad (t_{\rm evap} = t_{\rm universe} = 4.35 \times 10^{17} \text{ s})$$

UQFF Threshold

$$M_{\rm threshold}^{\rm UQFF} = M_{\rm threshold}^{\rm GR} \times (T_{\rm UQFF}/T_H)^{-4/3} = 5.70 \times 10^{11} \times 0.99^{-4/3} = 5.73 \times 10^{11} \text{ kg}$$

UQFF threshold: 5.73×10^{11} kg (vs GR 5.70×10^{11} kg) ■ 0.5% increase.

3. Present-Day PBH Gamma-Ray Flux

For PBHs near threshold ($M \sim M_{\rm threshold}$), the Hawking radiation contributes to the diffuse gamma-ray background:

$$\frac{d^2 \Phi_{\gamma}}{dE d\Omega} = \frac{1}{4\pi} \int_0^\infty n_{\rm PBH}(M) \frac{d^2 N_{\gamma}}{dE dt}(M, T_{\rm UQFF}) \, dM$$

The UQFF $T_{\rm UQFF} = 0.99 T_H$ reduces peak gamma-ray energy by 1%:

$$E_{\rm peak}^{\rm UQFF} = 2.82 k_B T_{\rm UQFF} = 2.82 k_B \times 0.99 T_H = 0.99 E_{\rm peak}^{\rm GR}$$

Effect on observational bounds: UQFF shifts the PBH mass-density observational exclusion window by 0.5% in M . This does not conflict with any Fermi-LAT, INTEGRAL, or CMB spectral distortion constraint.

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For $M_{\rm PBH}$ in the asteroid belt window ($10^5 \text{--} 10^7$ g, avoiding microlensing and CMB):

$$f_{\rm PBH}^{\rm UQFF} = f_{\rm PBH}^{\rm GR} \times \frac{M_{\rm threshold}^{\rm UQFF}}{M_{\rm threshold}^{\rm GR}} \times \frac{T_{\rm UQFF}^4}{T_H^4} = f_{\rm PBH} \times 1.005 \times 0.96 = f_{\rm PBH} \times 0.965$$

UQFF reduces $f_{\rm PBH}$ by 3.5% ■ within the 10–20% uncertainty on current PBH dark matter constraints.

Summary

PBH Property	Standard GR	UQFF	Change
Formation threshold δ_c	0.45	0.45 (unchanged)	$< 10^{-4}$
Survival mass threshold	$5.70 \times 10^7 \text{ kg}$	$5.73 \times 10^7 \text{ kg}$	+0.5%
Peak emission energy	E_{peak}	$0.99 E_{\text{peak}}$	-1%
f_{PBH} (dark matter)	f	$0.965f$	-3.5%
Fermi-LAT constraints	Not violated	Not violated	Compatible

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PBH Property	Standard GR	UQFF	Change
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UQFF Discovery: Novel application of UQFF calibration constants ($\beta = 5.0 \cdot 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. PBH Formation Threshold

Standard GR δ_c

$\delta_c = 0.45$; (Harrison-Zel'dovich radiation era)

UQFF Correction

The UQFF Buoyant mode adds vacuum pressure during the radiation era:

$$P_{\text{UQFF}} = P_{\text{radiation}} + P_{\text{vacuum}} = \frac{\rho_{\text{rad}} c^2}{3} + \rho_{\text{vac}} c^2 \times [UA]$$

The fractional correction:

$$\delta_c^{\text{UQFF}} = \delta_c \times \left(1 - \frac{P_{\text{vacuum}}}{P_{\text{rad}}}\right) = 0.45 \times (1 - [UA] \times z_{\text{formation}}^{-4})$$

At $z_{\text{formation}} = 106$ (typical PBH epoch): $P_{\text{vacuum}}/P_{\text{rad}} = 0.0001 \cdot 10^{-4} \approx$ negligible.

UQFF $\delta_c = 0.45$ (unchanged from GR) ■ the formation threshold is not perturbed.

2. Survival Mass Threshold

Standard GR

$$M_{\text{threshold}}^{\text{GR}} = 5.70 \times 10^{11} \text{ kg} \quad (t_{\text{evap}} = t_{\text{universe}} = 4.35 \times 10^{17} \text{ s})$$

UQFF Threshold

$$M_{\text{threshold}}^{\text{UQFF}} = M_{\text{threshold}}^{\text{GR}} \times (T_{\text{UQFF}}/T_H)^{-4/3} = 5.70 \times 10^{11} \times 0.99^{-4/3} = 5.73 \times 10^{11} \text{ kg}$$

UQFF threshold: $5.73 \cdot 10^{11}$ kg (vs GR $5.70 \cdot 10^{11}$ kg) ■ 0.5% increase.

3. Present-Day PBH Gamma-Ray Flux

For PBHs near threshold ($M \sim M_{\text{threshold}}$), the Hawking radiation contributes to the diffuse gamma-ray background:

$$\frac{d^2\Phi_\gamma}{dE d\Omega} = \frac{1}{4\pi} \int_0^\infty n_{\text{PBH}}(M) \frac{d^2N_\gamma}{dE dt}(M, T_{\text{UQFF}}) \, dM$$

The UQFF $T_{\text{UQFF}} = 0.99 T_H$ reduces peak gamma-ray energy by 1%:

$$E_{\text{peak}}^{\text{UQFF}} = 2.82 k_B T_{\text{UQFF}} = 2.82 k_B \times 0.99 T_H = 0.99 E_{\text{peak}}^{\text{GR}}$$

Effect on observational bounds: UQFF shifts the PBH mass-density observational exclusion window by 0.5% in M . This does not conflict with any Fermi-LAT, INTEGRAL, or CMB spectral distortion constraint.

4. PBH dark matter fraction f_{PBH}

For M_{PBH} in the asteroid belt window ($10^{15}\text{--}10^{17}$ g, avoiding microlensing and CMB):

$$\begin{aligned} f_{\text{PBH}}^{\text{UQFF}} &= f_{\text{PBH}}^{\text{GR}} \times \frac{M_{\text{threshold}}^{\text{UQFF}}}{M_{\text{threshold}}^{\text{GR}}} \times \frac{T_{\text{UQFF}}^4}{T_H^4} \\ &= f_{\text{PBH}} \times 1.005 \times 0.96 = f_{\text{PBH}} \times 0.965 \end{aligned}$$

UQFF reduces f_{PBH} by 3.5% within the 10–20% uncertainty on current PBH dark matter constraints.

Summary

PBH Property	Standard GR	UQFF	Change
Formation threshold dc	0.45	0.45 (unchanged)	$< 10^{-4}$
Survival mass threshold	5.70×10^{15} kg	5.73×10^{15} kg	+0.5%
Peak emission energy	E_{peak}	$0.99 E_{\text{peak}}$	-1%
f_{PBH} (dark matter)	f	$0.965f$	-3.5%
Fermi-LAT constraints	Not violated	Not violated	Compatible

Source: `validatehawkingtemperature.py` *primordial BH tests* | $\dot{N} = 0.0005/\text{day}$ | $[SSq] = 0.57$
.Groups[1].Value

Abstract

Primordial black holes (PBHs) form during radiation domination from density perturbations exceeding $dc \sim 0.45$. The UQFF modifies the critical density threshold dc_{UQFF} through the Buoyant vacuum pressure correction and extends PBH survival above the GR threshold mass $M_{\text{threshold}}$ by 3.5%. The evaporation temperature of surviving PBHs is $T_{\text{UQFF}} = 0.99 T_H$, slightly reducing their current gamma-ray flux. PBHs with $M \sim 10^{15}\text{--}10^{17}$ g may constitute part of dark matter. The UQFF predicts the dark matter fraction f_{PBH} is unaffected (the 3.5% mass threshold shift does not move PBHs into or out of the observationally allowed window).

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4$ day, $[SSq] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. PBH Formation Threshold

Standard GR dc

$$\Delta_c = 0.45 \text{ (Harrison-Zel'dovich radiation era)}$$

UQFF Correction

The UQFF Buoyant mode adds vacuum pressure during the radiation era:

$$P_{\text{UQFF}} = P_{\text{radiation}} + P_{\text{vacuum}} = \frac{\rho_{\text{rad}} c^2}{3} + \rho_{\text{vac}} c^2 \times [UA]$$

The fractional correction:

$$\Delta_c^{\text{UQFF}} = \Delta_c \times \left(1 - \frac{P_{\text{vacuum}}}{P_{\text{rad}}}\right) = 0.45 \times (1 - [UA] \times z_{\text{formation}}^{-4})$$

At $z_{\text{formation}} = 106$ (typical PBH epoch): $P_{\text{vacuum}}/P_{\text{rad}} = 0.0001 \times 10^4 \approx$ negligible.

UQFF dc = 0.45 (unchanged from GR) ■ the formation threshold is not perturbed.

2. Survival Mass Threshold

Standard GR

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$$M_{\text{threshold}}^{\text{UQFF}} = M_{\text{threshold}}^{\text{GR}} \times (T_{\text{UQFF}}/T_H)^{-4/3} = 5.70 \times 10^{11} \times 0.99^{-4/3} = 5.73 \times 10^{11} \text{ kg}$$

UQFF threshold: 5.73×10^{11} kg (vs GR 5.70×10^{11} kg) ■ 0.5% increase.

3. Present-Day PBH Gamma-Ray Flux

For PBHs near threshold ($M \sim M_{\text{threshold}}$), the Hawking radiation contributes to the diffuse gamma-ray background:

$$\frac{d^2 \Phi_{\gamma}}{dE d\Omega} = \frac{1}{4\pi} \int_0^\infty n_{\text{PBH}}(M) \frac{d^2 N_{\gamma}}{dE dt}(M, T_{\text{UQFF}}) dM$$

The UQFF $T_{\text{UQFF}} = 0.99 T_H$ reduces peak gamma-ray energy by 1%:

$$E_{\text{peak}}^{\text{UQFF}} = 2.82 k_B T_{\text{UQFF}} = 2.82 k_B \times 0.99 T_H = 0.99 E_{\text{peak}}^{\text{GR}}$$

Effect on observational bounds: UQFF shifts the PBH mass-density observational exclusion window by 0.5% in M . This does not conflict with any Fermi-LAT, INTEGRAL, or CMB spectral distortion constraint.

4. PBH dark matter fraction f_{PBH}

For M_{PBH} in the asteroid belt window ($10^{25}\text{--}10^{27}$ g, avoiding microlensing and CMB):

$$f_{\text{PBH}}^{\text{UQFF}} = f_{\text{PBH}}^{\text{GR}} \times \frac{M_{\text{threshold}}^{\text{UQFF}}}{M_{\text{threshold}}^{\text{GR}}} \times \frac{T_{\text{UQFF}}^4}{T_{\text{H}}^4}$$

$$= f_{\text{PBH}} \times 1.005 \times 0.96 = f_{\text{PBH}} \times 0.965$$

UQFF reduces f_{PBH} by 3.5% within the 10–20% uncertainty on current PBH dark matter constraints.

Summary

PBH Property	Standard GR	UQFF	Change
Formation threshold δ_{c}	0.45	0.45 (unchanged)	$< 10^{-4}$
Survival mass threshold	5.70×10^{25} kg	5.73×10^{25} kg	+0.5%
Peak emission energy	E_{peak}	$0.99 E_{\text{peak}}$	-1%
f_{PBH} (dark matter)	f	$0.965f$	-3.5%
Fermi-LAT constraints	Not violated	Not violated	Compatible

Source: `validatehawkingtemperature.py` *primordial BH tests* | $\dot{\gamma} = 0.0005/\text{day}$ | $[SSq] = 0.57$
`.Groups[1].Value "PAPER_0:D3" -f $n`

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Primordial black holes (PBHs) form during radiation domination from density perturbations exceeding $\delta_{\text{c}} \sim 0.45$. The UQFF modifies the critical density threshold δ_{cUQFF} through the *Buoyant vacuum pressure correction and extends PBH survival above the GR threshold mass $M_{\text{threshold}}$* by 3.5%. The evaporation temperature of surviving PBHs is $T_{\text{UQFF}} = 0.99 T_{\text{H}}$, slightly reducing their current gamma-ray flux. PBHs with $M \sim 10^{25}\text{--}10^{27}$ g may constitute part of dark matter ■ the UQFF predicts the dark matter fraction f_{PBH} is unaffected (the 3.5% mass threshold shift does not move PBHs into or out of the observationally allowed window).

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The UQFF Buoyant mode adds vacuum pressure during the radiation era:

$$P_{\rm UQFF} = P_{\rm radiation} + P_{\rm vacuum} = \frac{\rho_{\rm rad} c^2}{3} + \rho_{\rm vac} c^2 \times [UA]$$

The fractional correction:

$$\frac{\Delta P_{\rm UQFF}}{P_{\rm UQFF}} = \frac{\Delta P}{P} \times \left(1 - \frac{P_{\rm vacuum}}{P_{\rm rad}}\right) = 0.45 \times (1 - [UA] \times z_{\rm formation}^{-4})$$

At $z_{\rm formation} = 106$ (typical PBH epoch): $P_{\rm vacuum}/P_{\rm rad} = 0.0001 \ll 10^{-4}$? negligible.

UQFF dc = 0.45 (unchanged from GR) ■ the formation threshold is not perturbed.

2. Survival Mass Threshold

Standard GR

$$M_{\rm threshold}^{\rm GR} = 5.70 \times 10^{11} \text{ kg} \quad (t_{\rm evap} = t_{\rm universe} = 4.35 \times 10^{17} \text{ s})$$

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$$M_{\rm threshold}^{\rm UQFF} = M_{\rm threshold}^{\rm GR} \times (T_{\rm UQFF}/T_H)^{-4/3} = 5.70 \times 10^{11} \times 0.99^{-4/3} = 5.73 \times 10^{11} \text{ kg}$$

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The UQFF $T_{\rm UQFF} = 0.99 T_H$ reduces peak gamma-ray energy by 1%:

$$E_{\rm peak}^{\rm UQFF} = 2.82 k_B T_{\rm UQFF} = 2.82 k_B \times 0.99 T_H = 0.99 E_{\rm peak}^{\rm GR}$$

Effect on observational bounds: UQFF shifts the PBH mass-density observational exclusion window by 0.5% in M . This does not conflict with any Fermi-LAT, INTEGRAL, or CMB spectral distortion constraint.

4. PBH dark matter fraction $f_{\rm PBH}$

For $M_{\rm PBH}$ in the asteroid belt window ($10^5 \text{--} 10^7$ g, avoiding microlensing and CMB):

$$f_{\rm PBH}^{\rm UQFF} = f_{\rm PBH}^{\rm GR} \times \frac{M_{\rm threshold}^{\rm UQFF}}{M_{\rm threshold}^{\rm GR}} \times \frac{T_{\rm UQFF}^4}{T_H^4} = f_{\rm PBH} \times 1.005 \times 0.96 = f_{\rm PBH} \times 0.965$$

UQFF reduces $f_{\rm PBH}$ by 3.5% ■ within the 10–20% uncertainty on current PBH dark matter constraints.

Summary

PBH Property	Standard GR	UQFF	Change
Formation threshold α_c	0.45	0.45 (unchanged)	$< 10^{-4}$
Survival mass threshold	5.70×10^{22} kg	5.73×10^{22} kg	+0.5%
Peak emission energy	E_{peak}	$0.99 E_{\text{peak}}$	-1%
f_{PBH} (dark matter)	f	$0.965f$	-3.5%
Fermi-LAT constraints	Not violated	Not violated	Compatible

Source: `validatehawkingtemperature.py` primordial BH tests | $\gamma = 0.0005/\text{day}$ | $[SSq] = 0.57$

UQFF computed: Eddington luminosity UQFF correction = $1 - [SSq] \exp(-\gamma t) = 1 - 5.7 \times 10^{-4} \exp(-2.9 \times 10^{-4}) = 4.3 \times 10^{-1}$; F_U at event horizon = 2.0×10^{18} m/s.